2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to C*l*.

The ionic radius of Si is **not** shown.

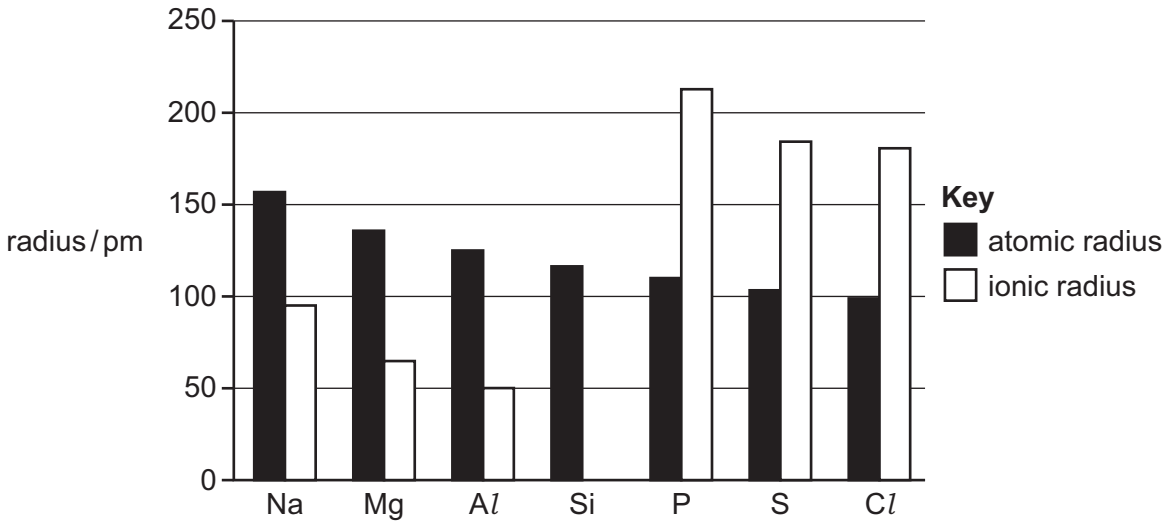


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to C*l*.

.....

.....

.....

..... [2]

(d) Group 2 nitrates decompose on heating to form oxides.

(i) State the trend in thermal stability of the Group 2 nitrates down the group.

..... [1]

(ii) Identify the other products of the thermal decomposition of Group 2 nitrates.

..... [1]

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) **(i)** Explain why the elements Na to A*l* are good electrical conductors.

.....

..... [1]

(iii) $S_2Cl_2(l)$ is yellow and $SCl_2(l)$ is red.

$$S_2Cl_2(l) + Cl_2(g) \rightleftharpoons 2SCl_2(l) \qquad \Delta H = -41\,kJ\,mol^{-1}$$

State what is observed when the following changes are made to an equilibrium mixture of $S_2Cl_2(l)$ and $SCl_2(l)$.

Explain your answers.

- The equilibrium mixture is warmed gently.

  observation .....

  explanation .....

  .....

  .....

- The overall pressure of the equilibrium mixture is increased.

  observation .....

  explanation .....

  .....

  .....

[4]

(f) Aqueous $MgCl_2$ reacts with aqueous Na_2CO_3 to form a white precipitate. Upon heating, the white precipitate undergoes thermal decomposition.

(i) Construct an equation to show the reaction of aqueous $MgCl_2$ with aqueous Na_2CO_3. Use state symbols in your equation.

..... [2]

(ii) Identify the products of the thermal decomposition of the white precipitate.

..... [1]

(iii) State the trend in thermal stability of the Group 2 carbonates down the group.

..... [1]

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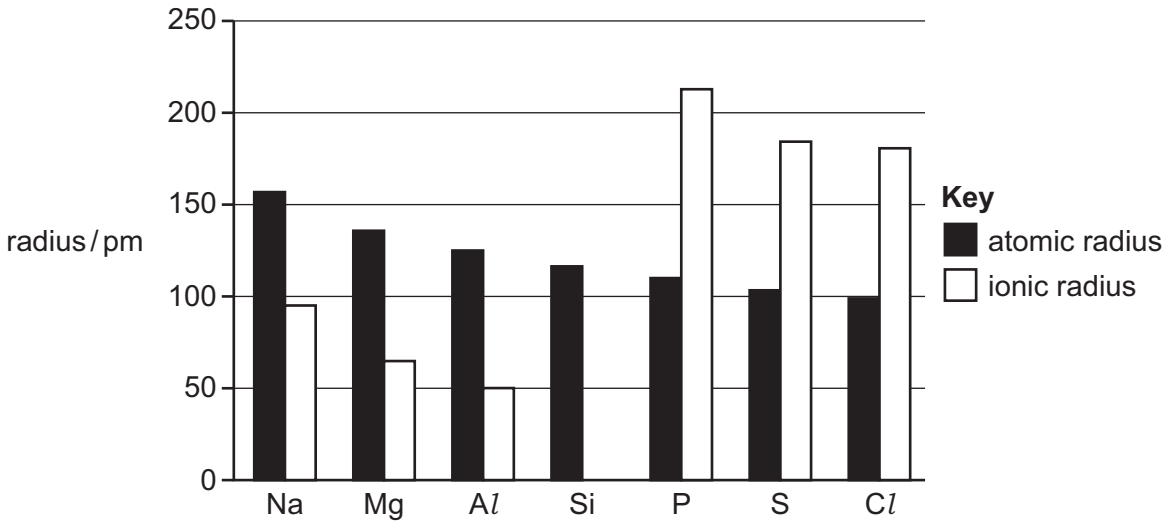


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.....

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(i) State the trend in thermal stability of the Group 2 nitrates down the group.

..... [1]

(ii) Identify the other products of the thermal decomposition of Group 2 nitrates.

..... [1]

2 **(a)** The Group 2 elements show metallic bonding.

Define metallic bonding.

.....

..... [2]

(b) The Group 2 elements form stable 2+ cations.

(i) State and explain the variation in ionic radius of the Group 2 elements down the group.

.....

.....

.....

..... [2]

(d) **(i)** State what is observed when dilute hydrochloric acid is added to separate samples of barium oxide and barium carbonate.

barium oxide .....

barium carbonate .....
[2]

(ii) Write an equation for the reaction of strontium, Sr, with an excess of cold water.

..... [1]

(iii) State the variation in solubility of the Group 2 sulfates down the group.

..... [1]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute HC*l*(aq).

..... [1]

(ii) Describe the thermal stability of the carbonates down Group 2.

..... [1]

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.

(i) Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural .....

man-made .....
[2]

(iv) Nitric acid reacts with basic oxides to form nitrates.

Write an equation for the reaction of nitric acid with calcium oxide.

..... [1]

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	C (graphite)	Sn
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(i) Write an equation for the reaction of magnesium carbonate with dilute HC*l*(aq).

..... [1]

1 **(a)** **(i)** Explain the lack of reactivity of nitrogen gas, $N_2(g)$.

.....

.....

..... [2]

(d) The oxide of silicon reacts with calcium oxide in an addition reaction to produce calcium silicate, $CaSiO_3$. The oxidation number of calcium in $CaSiO_3$ is +II.

(i) Deduce the oxidation number of silicon in calcium silicate.

oxidation number of silicon ..... [1]

(ii) Calcium oxide can be made from calcium carbonate in a single-step reaction.

Identify the type of reaction that occurs.

..... [1]

2 Separate samples of Na_2CO_3 and $NaHCO_3$ react with HCl(aq) to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	ΔH / kJ mol^{-1}
1	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$	ΔH_1
2	$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$	ΔH_2 = +27.2

(a) Complete the reaction pathway diagram in Fig. 2.1 for reaction 2.

Label the diagram to show the enthalpy change, ΔH_2, and the activation energy, E_A.

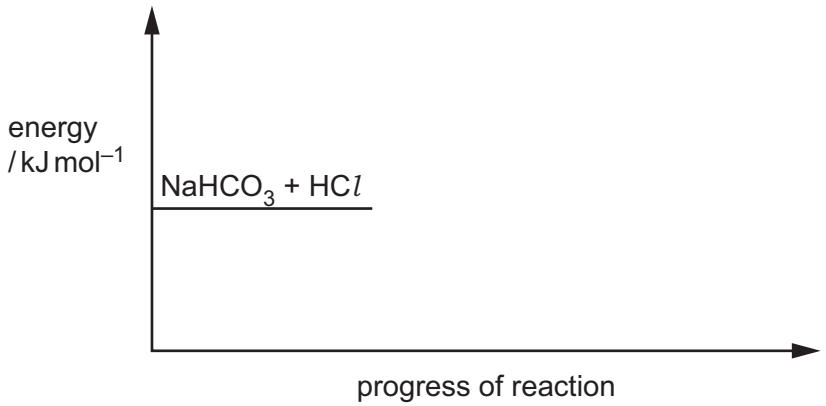


Fig. 2.1

[2]

(c) **Z** is a salt that contains a Period 4 element from Group 2. When **Z** is heated brown gas forms.

Identify the formula of **Z** and use it to write an equation for the reaction.

..... [2]

2 Barium hydroxide, $Ba(OH)_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $Ba(OH)_2$.

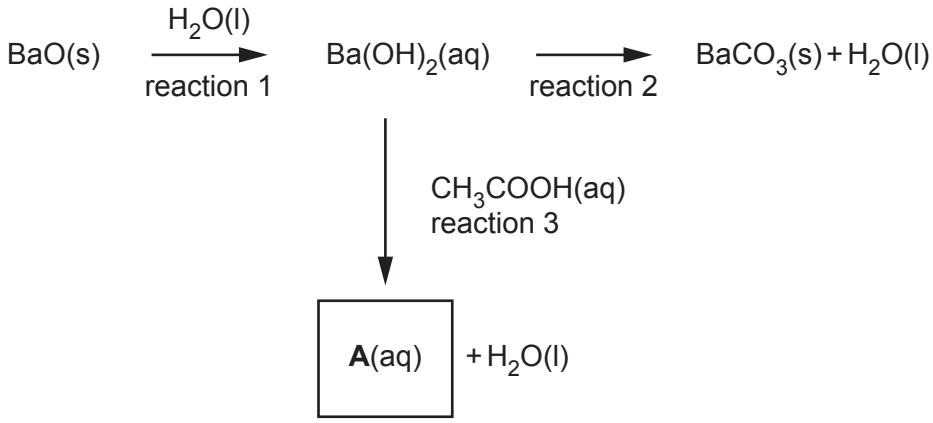


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

..... [1]

(ii) State what is observed in reaction 1.

..... [1]

(iii) Suggest a reactant for reaction 2.

..... [1]

(v) $Ba(OH)_2$ is made by the reaction of Ba with water.

Write an equation for this reaction.

..... [1]

(b) The mineral barytocalcite contains both $BaCO_3$ and $CaCO_3$. Both compounds decompose on heating.

(i) State which compound decomposes first when barytocalcite is heated.

Explain your answer.

.....

..... [1]

(ii) Construct an equation for the complete thermal decomposition of barytocalcite.

The formula of barytocalcite is $BaCa(CO_3)_2$.

$BaCa(CO_3)_2$ ..... [1]

2 NO and NO_2 react at 25 °C to give N_2O_3 as shown in the equation.

$$NO(g) + NO_2(g) \rightleftharpoons N_2O_3(g) \qquad \Delta H = -7.2\,kJ\,mol^{-1}$$

The reaction is reversible and reaches equilibrium in a closed system.

(a) Fig. 2.1 shows how the rate of the forward reaction changes with time.

Initially, the rate of the reverse reaction is zero.

Complete Fig. 2.1 to sketch how the rate of the **reverse** reaction changes with time.

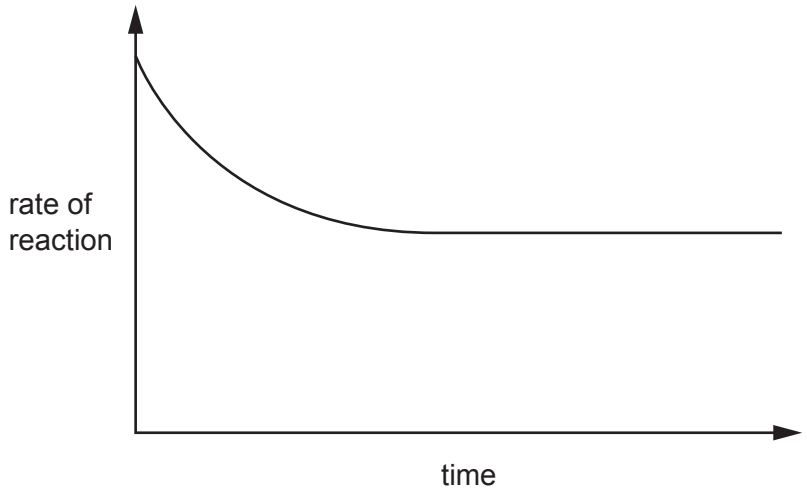


Fig. 2.1

[1]

(f) A student titrates nitric acid with a base to form a solution containing aqueous magnesium nitrate.

(i) Identify a base that the student could use.

..... [1]

(ii) The student evaporates the water to obtain magnesium nitrate solid. When this solid is heated it decomposes.

Write an equation for the decomposition of magnesium nitrate.

..... [1]

(iii) State how the thermal stability of Group 2 nitrates changes down the group.

..... [1]

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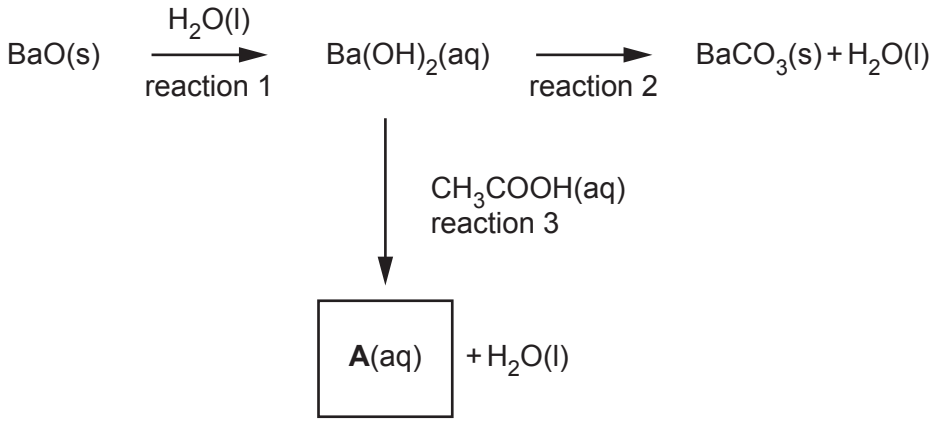


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Explain your answer.

.....

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The formula of barytocalcite is $BaCa(CO_3)_2$.

$BaCa(CO_3)_2$ ..... [1]

2 **(a)** The reaction of pure aluminium is only observed if the aluminium oxide layer is removed first. When pure aluminium is added to cold water, bubbles of gas are seen.

(i) State **one** property of aluminium oxide that explains why an aluminium object does **not** react with cold water until the aluminium oxide layer is removed.

..... [1]

(b) Aluminium nitrate is a white soluble salt. On heating aluminium nitrate, thermal decomposition occurs and a brown gas is seen.

State the formula of the salt of another element in Period 3 which also decomposes on heating to produce a brown gas.

..... [1]

2 A neutralisation reaction occurs when NaOH(aq) is added to H_2SO_4(aq).

equation 1 $2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

(a) Define enthalpy change of neutralisation, ΔH_{neut}.

.....

.....

..... [2]

(c) **(i)** Complete the equation for the reaction that occurs when a solution of $Ba(OH)_2$ is added to aqueous sulfuric acid. Include state symbols.

..... $H_2SO_4(aq)$ + ...$Ba(OH)_2(aq) \rightarrow$ ……
[2]

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(a) **(i)** Draw a labelled diagram to show the bonding and structure of the Group 2 metals at room temperature.

[2]

(b) Write an equation for the reaction of magnesium with cold water.

..... [1]

(c) Identify a single reagent that can be used to distinguish separate samples of dilute $Mg(NO_3)_2(aq)$ and dilute $Ba(NO_3)_2(aq)$.

Explain your answer.

reagent .....

explanation .....

.....

[2]